

COURSE TITLE : MAINTENANCE OF TEXTILE MACHINERIES
COURSE CODE : 6064
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

Module	Topics	Period
1	Introduction to maintenance, work shop machineries, bearings & lubricants.	15
2	Erection of textile machineries.	15
3	Maintenance of spinning machineries	15
4	Maintenance of weaving dyeing & printing machineries	15
TOTAL		60

COURSE OUTCOME

M0dule	Go	Student will be able to
1	1	Understand various maintenance operations in textile industry.
	2	Know a clear idea about the work shop machineries.
	3	Understand different lubricants & bearings used in textile industry.
2	1	Know the erection procedures for various textile machineries.
	2	Understand various types of various types of vibrations and balancing.
3	1	Know the maintenance schedule of different spinning machineries.
	2	Understand the various types of maintenance instruments.
	3	Know a clear idea about the various types of abrasives used in spinning mill and Safety precautions in textile mills.
4	1	Understand the various types of maintenance schedules of preparatory machines and looms.
	2	Know the maintenance schedule of different dyeing & printing machines.

SPECIFIC OUTCOME

MODULE I INTRODUCTION TO MAINTENANCE, WORK SHOP MACHINERIES, BEARINGS & LUBRICANTS.

1.1.0 Understand various maintenance operations in Textile Industry

1.1.1 State the objectives of maintenance.

1.1.2 List the various types of maintenance

1.1.3 Explain the Routine maintenance, Preventive maintenance, Predictive Maintenance, Planned, Emergency & Restorative maintenance

- 1.1.4 Illustrate the organization of maintenance department in a textile mill
- 1.1.5 List the duties & responsibilities of maintenance supervisor
- 1.1.6 Describe control of store, study of planning, scheduling and controlling of maintenance.
- 1.1.7 State the work order procedure, inventory card system & bin card system
- 1.1.8 Define work study techniques

1.2.0 Know a clear idea about the Work shop Machineries.

- 1.2.1 Describe various types of workshop machineries
- 1.2.2 List the different gauges & tools
- 1.2.3 Describe the process and objectives of Heat treatments
- 1.2.4 Explain the Special heat treatments for fluted roller, rings, spindles & calendar rollers
- 1.2.5 Describe the Protective coatings for metals

1.3.0 Understand different lubricants & bearings used in textile Industry.

- 1.3.1 Explain the different types of bearings.
- 1.3.2 State the functions of bearings, selection of bearings to textile machines
- 1.3.3 State the objectives of lubricants & characteristics of lubricants.
- 1.3.4 List the type of lubricants & lubrication
- 1.3.5 Illustrate the selection of lubricants & methods of application

MODULE II ERECTION OF TEXTILE MACHINERIES.

2.1.0 Know the erection procedures for various textile machineries

2.1.1 State the Importance of levelling & methods of levelling

- 2.1.2 List the Instruments used in levelling
- 2.1.3 Explain the erection procedures for various textile machineries – spinning, weaving and processing
- 2.1.4 Describe the metallic wire mounting – licker-in, cylinder, doffer & flat clipping
- 2.1.5 Explain the erection tools and equipments

2.2.0 Understand various types of various types of vibrations and Balancing

- 2.2.1 State the vibration of machines
- 2.2.2 Illustrate the causes of various types of vibrations
- 2.2.3 Describe the measurements of vibration
- 2.2.4 Illustrate the methods of damping
- 2.2.5 Describe the static and dynamic balancing

MODULE III MAINTENANCE OF SPINNING MACHINERIES

3.1.0 Know the maintenance schedule of different spinning machineries

- 3.1.1 Draw the chart of maintenance schedule of different spinning machineries for cleaning, Lubrication and repairs.
- 3.1.2 Explain the overhauling schedule of various spinning machineries
- 3.1.3 List the defects due to improper maintenance
- 3.1.4 Describe the maintenance schedule during strike & lay – off periods

3.2.0 Understand the various types of maintenance instruments

- 3.2.1 List the various types of maintenance instruments
- 3.2.2 Explain the ATIRA Roller tester, ATIRA NILO Meter and ATIRA TARP Gauge.
- 3.2.3 Describe the pneumafil suction pressure gauge'
- 3.2.4 Explain the tachometer, stroboscope

3.3.0 Know a clear idea about the variousTypes of abrasives used in spinning mill and Safety precautions in textile mills.

- 3.3.1 List the types of abrasives used in spinning mill & applications
- 3.3.2 Illustrate the roller cots – grinding – Buffing – Shore hardness treatments.
- 3.3.3 Describe the comber needling – needling for half lap.
- 3.3.4 Explain the safety precautions in textile mills & maintenance of safety equipments

MODULE IV MAINTENANCE OF WEAVING , DYEING & PRINTING MACHINERIES

4.1.0 Understand the various types of maintenance schedules of preparatory machines and looms

- 4.1.1 State the maintenance schedule for cleaning, lubrication, overhauling & settings
- 4.1.2 Explain the mmaintenance schedule for winding pirn winding, warping, sizing, plain & automatic looms
- 4.1.3 Identify the Care and maintenance of loom parts - heald – reed – picker- picking band – shuttle – buffer – temples - loom spindle – check strap
- 4.1.4 State the defects due to improper maintenance of preparatory machines and looms
- 4.1.5 List the causes of defects and rectification

4.2.0 Know the maintenance schedule of different Dyeing & printing machines

- 4.2.1 Describe the up keep of boilers and kiers.
- 4.2.2 Illustrate the maintenance schedule of padding mangles, stenter.
- 4.2.3 Prepare a maintenance schedule for dyeing machines.
- 4.2.4 Describe the maintenance schedule for printing machines
- 4.2.5 Explain the maintenance schedule for strikes and lay –off periods

CONTENT DETAILS

MODULE I INTRODUCTION TO MAINTENANCE, WORK SHOP MACHINERIES, BEARINGS & LUBRICANTS.

Objectives of maintenance- types of maintenance-Routine maintenance, Preventive maintenance, Predictive Maintenance, Planned, Emergency & Restorative maintenance - Organization of maintenance department in a textile mill - Duties & responsibilities of maintenance supervisor- Control of store-planning, scheduling and controlling of maintenance- Work order procedure, inventory card system& Bin card system - work study - techniques , objectives – Time study – motion study.

Work shop Machineries. Operations performed on milling, broaching, shaping ,grinding, drilling machines and lathe - gauges & tools , inside calliper , outside calliper, screw gauge, leaf gauge, depth

gauge - heat treatments –process and objectives –annealing , normalizing, hardening and tempering ,case hardening, carburizing, cyaniding, nitriding - special heat treatments for fluted rollers - protective coatings for metals.

Lubricants & bearings -Definition and types of bearings- solid journal ,split ,needle, roller and ball bearings - selection of bearings to textile machines - objectives of lubricants - characteristics of lubricants- type of lubricants –solid, semisolid and liquid lubricants with examples - selection of lubricants.

MODULE II ERECTION OF TEXTILE MACHINERIES.

Levelling - methods and objectives of levelling - Instruments used in levelling – spirit level, plumb line, wedge and spirit level , protractor - erection procedures for various textile machineries –erection of card , draw frame , simplex, ring frame and loom - metallic wire mounting – licker-in, cylinder, doffer & flat clipping - erection tools and equipments.

Vibration and Balancing

Vibration – definition and types, free vibration and forced vibration - Causes of various types of vibrations - measurement of vibration - damping of vibration – balancing – definition and types - static and dynamic balancing.

MODULE III MAINTENANCE OF SPINNING MACHINERIES

Maintenance schedule of different spinning machineries – Schedule for blow room, carding , drawing , lap formers , comber, roving , ring frame , ring doubler and TFO - overhauling schedule of various spinning machineries - defects due to improper maintenance - maintenance schedule during strike & lay – off periods

Maintenance instruments

Maintenance instruments - ATIRA Roller tester, ATIRA NILO meter and ATIRA TARP Gauge-pneumafil suction pressure gauge- Tachometer – Stroboscope

Abrasives and Safety precautions in textile mills.

Types of abrasives used in spinning mill – applications- cots – Buffing – Shore hardness treatments- comber needling – needling for half lap- safety precautions in textile mills -maintenance of safety equipments.

MODULE IV MAINTENANCE OF WEAVING , DYEING & PRINTING MACHINERIES

Maintenance schedule for Cleaning, Lubrication, Overhauling & Settings -maintenance schedule for Winding, Pirn winding, Warping, Sizing , Plain & automatic Looms - Care and maintenance of loom parts - Heald, Reed, Picker, Picking band , Shuttle, Buffer, Temples, box spindle and check strap - defects due to improper maintenance of preparatory machines and looms - List the causes of defects and rectification.

Maintenance schedule of different Dyeing & printing machines

Up keep of Boilers and Kiers- maintenance schedule of Padding mangles, Stenter-maintenance schedule for Dyeing machines- maintenance schedule for printing machines-maintenance schedule for strikes and lay –off periods

REFERENCE BOOKS

1. T V Rathinam -. Maintenance management in spinning - SITRA
2. Maintenance of textile machineries (Spinning, Weaving & Processing) - Tairo Publications
3. J.B.Joshi , K.B Vora - Spinning, Weaving & Processing machineries – Wood head publishing Ltd.,
4. ATIRA Process control in weaving - ATIRA
5. A r Garde - Norms for mechanical processing - BITRA
6. A T Shahani - Maintenance in ring frame - BITRA
7. ATIRA Maintenance in chemical processing department - ATIRA
8. Repair and adjustment of Textile machineries - M.R.Publications
9. BITRA Maintenance schedule, practice and
Check – points in spinning - BITRA
10. R.S Khurmi ,J.K.Guptha - Workshop Technology -
11. R.S.Khurana- R.S.Sedha. Material Science –
12. ATIRA Process control in Spinning – ATIRA